Experiment 8

Student Name : Sejal
UID: 24BAI70050
Branch: CSE - AIML
Section/Group: 24AIT_KRG G1
Semester: 4
Subject Code: 24CSH-298
Subject Name: DBMS

Aim

To create and execute a PL/pgSQL stored procedure that counts the number of employees based on gender using input, output, and inout parameters.

Software Requirements

- PostgreSQL (version 10 or above)
- pgAdmin / SQL Shell (psql)
- Operating System: Windows

Objectives

- To understand the concept of stored procedures in PL/pgSQL
- To learn the use of IN, OUT, and INOUT parameters
- To perform data retrieval using SELECT INTO
- To execute procedures using the CALL statement
- To display output using RAISE NOTICE

Practical/Experiment Steps

- Create a table employees with required fields.
- Insert sample data into the table.
- Create a stored procedure using PL/pgSQL.
- Use input parameter to pass gender.
- Use output parameter to store count.
- Use INOUT parameter to track status.
- Call the procedure inside a DO block.
- Display results using RAISE NOTICE.

Procedure

1. Table Creation

- A table employees is created with:
  - emp_id → Primary key
  - emp_name → Employee name
  - gender → Gender of employee
  - salary → Salary

2. Data Insertion

- Sample employee records are inserted into the table.

3. Stored Procedure Creation

- Procedure name: get_Employee_Count_BY_Gender
- Parameters:
  - IN_GENDER → Input gender
  - OUT_COUNT → Stores employee count
  - STATUS → Stores execution status

4. Logic Used

- SELECT COUNT(*) INTO OUT_COUNT counts employees.
- STATUS := 'SUCCESS' updates execution status.

5. Execution Block

- A DO block is used to:
  - Declare variables
  - Call the procedure
  - Print output using RAISE NOTICE

Input/Output Analysis

```
CREATE TABLE employees (
   emp_id SERIAL PRIMARY KEY,
   emp_name VARCHAR(50),
   gender VARCHAR(10),
   salary NUMERIC(10,2)
);
```

Output

```
Data Output   Messages   Notifications

CREATE TABLE

Query returned successfully in 478 msec.
```

INSERT INTO employees (emp_name, gender, salary) VALUES
('Amit', 'Male', 30000),
('Riya', 'Female', 35000),
('John', 'Male', 28000),
('Sneha', 'Female', 40000),
('Rahul', 'Male', 32000);

Output

```
Data Output   Messages   Notifications

INSERT 0 5

Query returned successfully in 167 msec.
```

CREATE OR REPLACE PROCEDURE get_Employee_Count_BY_Gender (IN IN_GENDER VARCHAR(20), OUT OUT_COUNT INT, INOUT STATUS VARCHAR(20))
AS
$$
BEGIN

SELECT COUNT(*) INTO OUT_COUNT FROM employees WHERE GENDER='Male';
STATUS:='SUCCESS';

END;

$$ LANGUAGE PLPGSQL;

Output

```
Data Output   Messages   Notifications

CREATE PROCEDURE

Query returned successfully in 254 msec.
```

```
DO
$$
DECLARE
GEN VARCHAR(20):='Male';
Count_of_Employee int;
STATUS VARCHAR:='Pending';
BEGIN
	CALL  get_Employee_Count_BY_Gender(GEN,Count_of_Employee,STATUS);
	RAISE NOTICE 'YOUR COUNT OF EMPLOYEE BY GENDER IS % AND YOUR STATUS IS %',Count_of_Employee,STATUS;
END;

$$
```

Output

```
Data Output   Messages   Notifications

NOTICE:  YOUR COUNT OF EMPLOYEE BY GENDER IS 3 AND YOUR STATUS IS SUCCESS
DO

Query returned successfully in 134 msec.
```

Learning Outcomes

- Understand how to create **stored procedures in PostgreSQL**
- Use **IN, OUT, INOUT parameters effectively**
- Perform **data retrieval using SELECT INTO**
- Execute procedures using CALL
- Handle procedural logic in PL/pgSQL